

KIRAT BAINS

Backend Developer

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SUMMARY

Backend Developer with experience in Go and Python, specializing in secure API design, microservices, and hybrid database solutions (Neo4j + PostgreSQL). Skilled in integrating AI/ML models (Hugging Face, Azure Cognitive Services) for personalization and automation. Strong background in cloud-native deployments (GCP, DigitalOcean, Azure), containerization (Docker), and performance optimization. Proven track record in workflow automation, OCR pipelines, and real-time recommendation systems that improve efficiency and scalability.

PROFESSIONAL

Backend Developer | Metric Design | Jan 2025 – Present

- Architected and implemented a scalable backend in Go, leveraging modular handler design, middleware, and session management using SCS for secure user workflows.
- Integrated Neo4j and PostgreSQL in a hybrid database model, ensuring transactional integrity between graph-based recommendations and relational user data.
- Developed recommendation logic using embedding-based similarity search, incorporating Hugging Face models and Moondream for preference mapping and AI-driven personalization.
- Built secure REST endpoints with CSRF protection, role-based access control (RBAC), and data validation to meet OWASP security guidelines.
- Implemented product ingestion and synchronization pipelines, ensuring real-time consistency between SQL and Neo4j datasets.
- Designed and deployed containerized microservices on DigitalOcean using Docker, with internal networking secured via Tailscale VPN for encrypted communication.
- Improved system performance and maintainability by optimizing database queries and implementing structured logging and error handling for production readiness.

Software Developer | Northam Distributors | Apr 2024 – Dec 2024

- Developed an event-driven document automation system using Python, GCP Eventarc, and Cloud Vision OCR to extract and rename large PDF files from Google Drive, reducing manual processing time by 75%.
- Deployed containerized services on GCP Cloud Run, enabling scalability and cost-efficient serverless execution across multiple client workflows.
- Implemented secure secret management with Google Cloud Secret Manager to protect API keys and credentials, ensuring compliance with security best practices.
- Configured centralized logging with Google Cloud Logging, improving observability, error tracking, and operational reliability in production environments.

Backend Developer | Dicta | May 2023 – Apr 2024

- Developed a cloud-based Text-to-Speech API using Azure Cognitive Services and GCP Storage, enabling efficient audio generation from large text files.
- Implemented an input processing pipeline to extract and process data from GCP buckets, splitting large text into smaller segments for seamless API performance.
- Integrated Azure Machine Learning speech models, improving processing efficiency by 40% while maintaining high-quality voice synthesis.
- Containerized the application with Docker for cross-platform portability and simplified cloud deployment.
- Improved system reliability by designing structured workflows, separating output and error files into dedicated GCP buckets for organized retrieval.

PROJECTS

Algorithmic Trading & Market Prediction

- Building an *AI-driven* trading platform using Python, and *deep learning* (LSTM, Transformers) for price forecasting, integrating real-time market data APIs, sentiment analysis (Hugging Face NLP), and deploying via FastAPI, Docker, and Cloud Run for production-ready trading automation.

Fantasy Manager

- Developed a fantasy football analytics MVP with Go backend and Python FastAPI microservice, PostgreSQL for persistent storage, and Redis for caching. Implemented trade valuation algorithms, real-time player updates via API integration, and containerized deployment with Docker Compose.

Lush POS System

- Developed a mobile-friendly *Point-of-Sale* web app using React for in-store product scanning, checkout, and inventory management, enabling employees to process sales on handheld devices.

E-Commerce Web Application

- Built a *full-stack* platform using React, Redux, Node.js, Express, and MySQL with features like user authentication, product catalog, shopping cart, and order management, deployed with Knex-based migrations and secure JWT auth.

TECHNICAL SKILLS

Languages:

- Python
- Golang
- SQL / CQL
- HTML / CSS
- NodeJS / React

Frameworks:

- FastAPI
- Flask
- Express JS
- Chi

Databases:

- Neo4J
- PostgreSQL
- MySQL
- Prometheus

GCP:

- Cloud Run
- Pub/Sub
- Secret Manager
- Cloud Logging
- Cloud Vision API

Machine Learning:

- Hugging Face
- Moondream AI
- CLIP
- ControlNet

DevOps:

- Postman
- Docker / Podman
- Tailscale
- ControlNet

Security:

- CSRF
- RBAC
- SCS
- IAM

EDUCATION & CERTIFICATIONS

Bachelor of Arts (Philosophy) – University of British Columbia

Diploma in Software Engineering – BrainStation Vancouver

Certifications: GCP Associated Cloud Engineer, Azure AZ-900